

# USB2RIG Wiring

by Cam VE3CA

The USB2RIG is an economical audio and serial interface to connect ham radios to computers for digital programs on both HF and VHF. It is similar to the Digirig with a CM108 chip and available on Amazon.

The USB2RIG has two 3.5mm 4 pin connectors, one is for Audio and PTT, the other is for serial rig control.



**NOTE:** These connectors are only 7mm apart, making it difficult to have both in at one time without modification of the connectors.



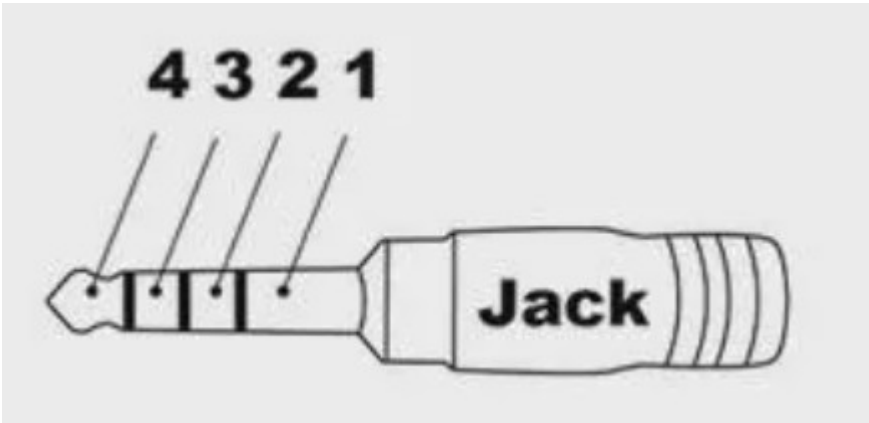
# Solving the Connector Problem

If rig control is not needed, you only need to use the Audio connector, which has Audio and PTT, so there is no problem. If you want rig control, it is possible to fit two plugs in without covers. In this case you could use some shrink wrap. instead



# Default Wiring

The pins are commonly numbered from shaft to tip as shown below.

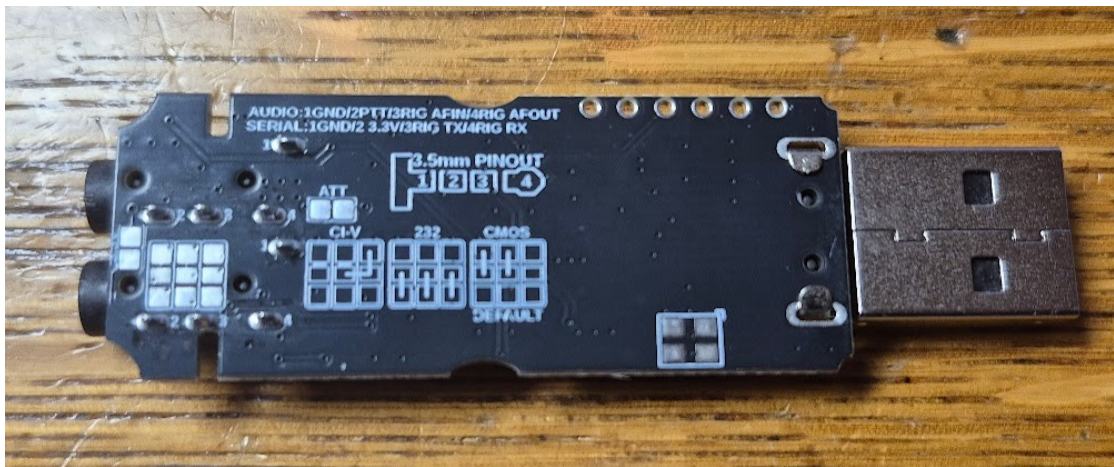
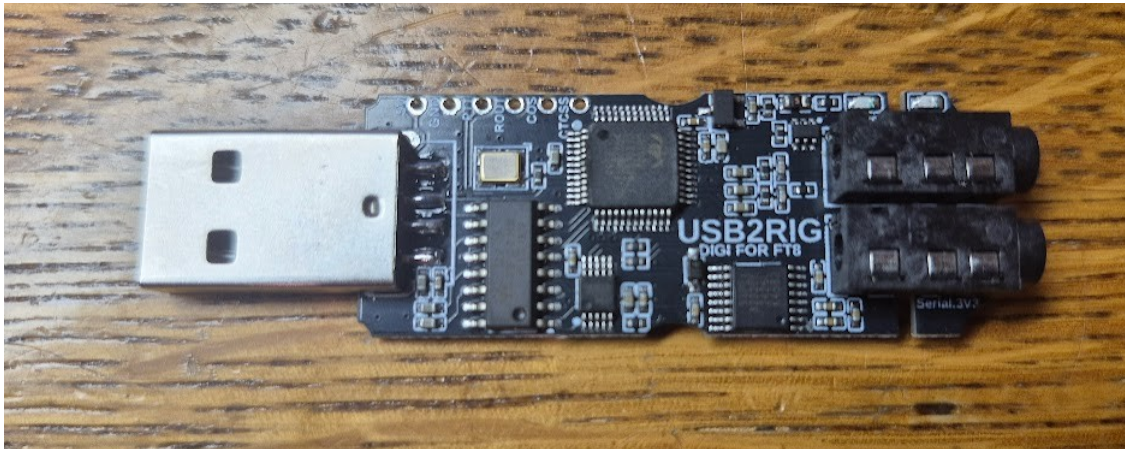


| Pin       | Serial Connector Usage (default) | Audio Connector Usage |
|-----------|----------------------------------|-----------------------|
| 1 (Shaft) | Ground                           | Ground                |
| 2 (Ring)  | 3.3V (not connected)             | PTT (open collector)  |
| 3 (Ring)  | RXD (from rig)                   | TX Audio (to rig)     |
| 4 (Tip)   | TXD (to rig)                     | RX Audio (from rig)   |

By default, the serial port is wired for TTL levels (5V/0V non-inverted) and the 3.3V ring is not connected. The configuration can be changed by opening the dongle and changing jumpers.

# Changing Configuration

Opening the dongle is tricky. It can be heated with a hair dryer and pried apart. Inside, the board looks like this.



There is a small trace that would have to be cut to undo the default pad connections in the upper left and upper centre of the 9 pad block. Then you can add jumpers for the CI-V or RS232 configuration.

To enable 3.3V to pin 2 of the serial connector, you need to add a jumper to the two pads at the far left at the edge of the board.